

ILATE

Composite func = By Parts.
 $\int_I f(n) g(n) dn$

$$\int u dv = uv - \int v du$$

⇒ Integration by Parts

Evaluate the integrals in Exercise 1-24, using integration by Parts.

01) $\int n \sin \frac{n}{2} dn$

First simple, Second integrate
 $\int$ Derivate of First, Second integrate
 double integrate

Solution:

Let: $u = n$	$du = 1 dn$
$dv = \sin \frac{n}{2}$	$v =$ cancel

✓ Remember always.

$f(n) g(n)$

First Simple ($f(n)$)

Second integrate ($g(n)$)

$\int$ Derivative of First ($f(n)$)
 Second integrate ($g(n)$)
 Double integrate.

✓ Let $f(n) = n$

$g(n) = \sin \frac{n}{2}$

$$= n \int \sin \frac{n}{2} dn - \int \left(\frac{d(n)}{dn} \cdot \int \left(\sin \frac{n}{2} dn \right) \right) dn$$

$$= n \left(\frac{-\cos \frac{n}{2}}{1/2} \right) - \int \left(1 \cdot \left(\frac{-\cos \frac{n}{2}}{1/2} \right) \right) dn$$

$$= 2 \cdot \frac{-n \cos \frac{n}{2}}{2} + \int \frac{2 \cos \frac{n}{2}}{2} dn$$

$$= \frac{-2n \cos \frac{n}{2}}{2} + 2 \int \cos \frac{n}{2} dn$$

angle = $\frac{1}{2}$

$$= \frac{-2n \cos \frac{n}{2}}{2} + 2 \left(\frac{\sin \frac{n}{2}}{1/2} \right) + C$$

$$= \frac{-2n \cos \frac{n}{2}}{2} + 2 \cdot \frac{1}{2} \frac{\sin n}{2} + C$$

$$\boxed{\frac{-2n \cos \frac{n}{2}}{2} + 4 \frac{\sin n}{2} + C}$$

Answer,

Formula, $\int f(x) \cdot g(x) dx = f(x) \int g(x) dx - \int f'(x) \cdot \int g(x) dx$

02) $\int \theta \cos \pi \theta d\theta$

Solution: Let:-

$f(n) = \theta$

$g(n) = \cos \pi \theta$

$$= \theta \int \cos \pi \theta d\theta - \int \left(\frac{d}{d\theta} (\theta) \cdot \int \cos \pi \theta d\theta \right) d\theta$$

$$= \theta \left(\frac{\sin \pi \theta}{\pi} \right) - \int 1 \cdot \left(\frac{\sin \pi \theta}{\pi} \right) d\theta$$

$$= \frac{\theta}{\pi} \sin \pi \theta - \frac{1}{\pi} \int \sin \pi \theta d\theta$$

$$= \frac{\theta}{\pi} \sin \pi \theta - \frac{1}{\pi} \left(-\frac{\cos \pi \theta}{\pi} \right) + C$$

$$= \left[\frac{\theta}{\pi} \sin \pi \theta + \frac{1}{\pi^2} \cos \pi \theta + C \right] \text{Ans.}$$

03) $\int t^2 \cos t dt$

Sol:-

Let, $f(n) = t^2$ $g(n) = \cos t$

$$= f(n) \int g(n) dt - \int \left(f'(n) \cdot \int g(n) dt \right) dt$$

$$= t^2 \int \cos t dt - \int \left(\frac{d}{dt} (t^2) \cdot \int \cos t dt \right) dt$$

$$= t^2 \left(\frac{\sin t}{1} \right) - \int \left(2t \cdot \frac{\sin t}{1} \right) dt$$

$t^2 \sin t - 2 \int (t \cdot \sin t) dt$ Again by parts at that. Let, $f(n) = t$, $g(n) = \sin$

$$t^2 \sin t - 2 \int f(n) \cdot g(n) dt - \int f'(n) \cdot \int g(n) dt$$

$$t^2 \sin t - 2 \left[(t) (-\cos t) - \int (1) \cdot (-\cos t) dt \right]$$

$$t^2 \sin t - 2 \left(-t \cos t + \sin t \right) + C$$

$$\boxed{t^2 \sin t + 2t \cos t - 2 \sin t + C} \text{ Answer}$$

Q4) $\int n^2 \sin n \, dn$

Sol:- let, $f(n) = n^2$, and $g(n) = \sin n$

$$= f(n) \int g(n) \, dn - \int f'(n) \int g(n) \, dn$$

$$n^2 \int \sin n \, dn - \int \frac{d}{dn} (n^2) \int \sin n \, dn$$

$$n^2 \cdot -\cos n - \int (2n \cdot -\cos n) \, dn$$

$$n^2 \cdot -\cos n - 2 \int n \cos n \, dn$$

$$-n^2 \cos n - 2 \int n \cos n \, dn$$

$$-n^2 \cos n + 2 \left[\int n \cos n \, dn \right] \text{ Again Integration by Parts}$$

Let, $f(n) = n$, $g(n) = \cos n$

$$-n^2 \cos n + 2 \left[f(n) \int g(n) \, dn - \int f'(n) \cdot g(n) \, dn \right]$$

$$-n^2 \cos n + 2 \left[n \int \cos n \, dn - \int \frac{d}{dn} (n) \cdot \cos n \, dn \right]$$

$$-n^2 \cos n + 2 \left[n \int (\cos n) \, dn - \int (1) \cdot \cos n \, dn \right]$$

$$-n^2 \cos n + 2n \sin n - 2 \int \cos n \, dn$$

$$-n^2 \cos n + 2n \sin n - 2 \cdot (-\cos n) + C$$

$$\boxed{-n^2 \cos n + 2n \sin n + 2 \cos n + C} \text{ Answer}$$

$$Q5) \int_1^2 x \ln x \, dx$$

Let, $f(x) = \ln x$, and $g(x) = x$

$$f(x) \int g(x) \, dx - \int \frac{d}{dx} (f(x)) \cdot \int g(x) \, dx$$

$$\ln x \int x \, dx - \int \left(\frac{d}{dx} (\ln x) \cdot \int x \, dx \right) dx$$

$$= \ln x \cdot \left[\frac{x^2}{2} \right]_1^2 - \int \left(\frac{1}{x} \cdot \frac{x^2}{2} \right) dx$$

$$\left[\ln x \cdot \frac{x^2}{2} \right]_1^2 - \int \left(\frac{1}{2} \right) dx$$

$$\left[\ln x \cdot \frac{x^2}{2} \right]_1^2 - \frac{1}{2} \int x \, dx$$

$$\left[\ln x \cdot \frac{x^2}{2} - \frac{1}{2} \cdot \frac{x^2}{2} \right]_1^2$$

Putting limits = upper = $\ln(2) \cdot \frac{2^2}{2}$ lower = $\ln(1) \cdot \frac{(1)^2}{2}$

$$= \frac{2^2}{2} \ln(2) - \frac{1}{2} \cdot \frac{(1)^2}{2}$$

$$\left[\ln(2) \cdot \frac{4}{2} - \ln(1) \cdot \frac{1}{2} \right] - \frac{1}{2} \cdot \frac{4}{2} + \frac{1}{2} \cdot \frac{1}{2}$$

$$= \frac{4 \ln(2)}{2} - \frac{1}{4} + \frac{1}{4}$$

$$= \frac{4 \ln(2)}{2} = 2 \ln(2)$$

Formula: $\int u dv = f(u) \int g(u) - \int (f'(u) \cdot \int g(u) du) du$

Q6) $\int_1^e n^3 \ln n \, dn$

For this

Solution: ILATE $\Rightarrow f(u) = \ln n$, $g(u) = n^3$

= we can rewrite this as: (order-wise)

$\int_1^e \ln n \cdot n^3 \, dn$ → Question Rewriting...

$f(n) = \ln n$, and $g(n) = n^3$

$= \ln n \int_1^e n^3 \, dn - \int_1^e \left(\frac{d}{dn} \ln n \cdot \int_1^e n^3 \, dn \right) dn$

of Formula

No, Because we write whole integral

$= \ln n \cdot \frac{n^4}{4} \Big|_1^e - \int_1^e \left(\frac{1}{n} \cdot \frac{n^4}{4} \right) dn$

$\frac{1}{4} \ln n \cdot n^4 \Big|_1^e - \frac{1}{4} \int_1^e n^3$

$\frac{1}{4} \ln n \cdot n^4 \Big|_1^e - \frac{1}{4} \cdot \frac{n^4}{4} \Big|_1^e$

Putting limit

upper = $\frac{1}{4} \ln(e) \cdot e^4$

$-\frac{1}{16} \cdot n^4 \Big|_1^e$

→ Putting upper lower limit

lower = $\frac{1}{4} \ln(1) \cdot (1)^4$

$-\frac{1}{16} [e^4 - 1^4]$ → combine both.

$\frac{1}{4} (1) \cdot e^4 - (-0)$

$\frac{1}{4} e^4 - \frac{1}{16} e^4 + \frac{1}{16}$

$\frac{4e^4 - e^4 + 1}{16} = \frac{3e^4 + 1}{16}$

ILATE

$$\int e^{3n} = \frac{e^{3n}}{3(1)} \rightarrow \text{down derivative of power}$$

L_{Ans}

07) $\int n e^n dn$

Solution:

$f(n) = n$, and $g(n) = e^n$

$$= f(n) \int g(n) dn - \int \left(f'(n) \cdot \int g(n) dn \right) dn.$$

$$= n \cdot \int e^n dn - \int \left(\frac{d}{dn} (n) \cdot \int e^n dn \right) dn.$$

$$= n \cdot e^n - \int (1) \cdot e^n dn$$

$$n e^n - \int (e^n dn)$$

$$n e^n - e^n + C$$
$$= \boxed{n e^n - e^n + C} \text{ Answer.}$$

08) $\int n e^{3n} dn$

Solution: Let, $f(n) = n$, and $g(n) = e^{3n}$

$$= f(n) \int g(n) dn - \int \left(f'(n) \cdot \int g(n) dn \right) dn$$

$$n \int e^{3n} dn - \int \left(\frac{d}{dn} (n) \cdot \int e^{3n} dn \right) dn$$

$$n \cdot \frac{e^{3n}}{3(1)} - \int (1) \cdot \frac{e^{3n}}{3(1)} dn$$

$$\frac{n \cdot e^{3n}}{3} - \int \left(\frac{e^{3n}}{3} \right) dn$$

$$\frac{n \cdot e^{3n}}{3} - \frac{1}{3} \int (e^{3n}) dn$$

$$\frac{n \cdot e^{3n}}{3} - \frac{1}{3} \cdot \frac{e^{3n}}{3} + C$$

$$\boxed{\frac{n \cdot e^{3n}}{3} - \frac{1}{9} e^{3n} + C} \text{ Answer}$$

09) $\int n^2 e^{-n} dn$

Solution:

$f(n) = n^2$ and $g(n) = e^{-n}$

$$f(n) \int g(n) dn - \int (f'(n) \cdot \int g(n) dn) dn$$

$$n^2 \cdot \int e^{-n} dn - \int \left(\frac{d}{dn} (n^2) \cdot \int e^{-n} dn \right) dn$$

$$n^2 \cdot \frac{e^{-n}}{-1} - \int \left(2n \cdot \frac{e^{-n}}{-1} \right) dn$$

$$-n^2 e^{-n} - \int (-2n \cdot e^{-n}) dn$$

$$-n^2 e^{-n} + 2 \int (n e^{-n}) dn$$

Again Integration by Parts $f(n) = n$, $g(n) = e^{-n}$

$$= -n^2 e^{-n} + 2 \left[f(n) \int g(n) dn - \int f'(n) \cdot \int g(n) dn \right]$$

$$= -n^2 e^{-n} + 2 \left[(n) \int e^{-n} dn - \int \left(\frac{d}{dn} (n) \cdot \int e^{-n} dn \right) dn \right]$$

$$= -n^2 e^{-n} + 2 \cdot \left[\begin{array}{c} n \cdot \frac{e^{-n}}{-1} \\ \text{Multipl} \end{array} - 2 \int \left((1) \cdot \frac{e^{-n}}{-1} \right) dn \right]$$

$$= -n^2 e^{-n} - 2n e^{-n} + 2 \int e^{-n} dn$$

$$= -n^2 e^{-n} - 2n e^{-n} + 2 \cdot \frac{e^{-n}}{-1} + c$$

$$= -n^2 e^{-n} - 2n e^{-n} - 2e^{-n} + c$$

Answer →

$$= \boxed{-n^2 e^{-n} - 2n e^{-n} - 2e^{-n} + c} \quad \text{Ans}$$

ILATE

$$10) \int (n^2 - 2n + 1) e^{2n} dn$$

Solution:-

$$f(n) = (n^2 - 2n + 1) \text{ and } g(n) = e^{2n}$$

$$= f(n) \int g(n) dn - \int (f'(n) \cdot \int g(n) dn) dn$$

$$= (n^2 - 2n + 1) \int e^{2n} dn - \int \left[\left(\frac{d}{dn} n^2 - 2n + 1 \right) \cdot \int e^{2n} dn \right] dn$$

$$= (n^2 - 2n + 1) \cdot \frac{e^{2n}}{2} - \int \left[(2n - 2) \cdot \frac{e^{2n}}{2} \right] dn$$

$$= (n^2 - 2n + 1) \cdot \frac{e^{2n}}{2} - \frac{1}{2} \int (2n - 2) \cdot (e^{2n}) dn$$

$f(n) = (2n - 2)$
Again Integration by parts $g(n) = (e^{2n})$

$$(n^2 - 2n + 1) \cdot \frac{e^{2n}}{2} - \frac{1}{2} \left[f(n) \int g(n) dn - \int (f'(n) \cdot \int g(n) dn) dn \right]$$

$$(n^2 - 2n + 1) \cdot \frac{e^{2n}}{2} - \frac{1}{2} \left[(2n - 2) \int e^{2n} dn - \int (2) \cdot \int e^{2n} dn \right] dn$$

$$(n^2 - 2n + 1) \cdot \frac{e^{2n}}{2} - \frac{1}{2} \left[2n - 2 \cdot \frac{e^{2n}}{2} - \int \left(\frac{2e^{2n}}{2} \right) dn \right]$$

$$(n^2 - 2n + 1) \cdot \frac{e^{2n}}{2} - \frac{1}{4} \left[2n - 2 \cdot e^{2n} + 1 \int (e^{2n}) dn \right]$$

$$\frac{n^2 - 2n + 1 \cdot e^{2n}}{2} - \frac{1}{4} \left[2n - 2 \cdot e^{2n} + 1 \cdot \frac{e^{2n}}{2} \right] + C$$

$$\left(n^2 - 2n + 1 \right) \frac{e^{2n}}{2} - \frac{1}{4} 2n - 2e^{2n} + \frac{1}{4} e^{2n} + C$$

Answer

ILATE
↳ First → inverse

11) $\int \tan^{-1} y \, dy$

Solution:-

$$\int y^0 \tan^{-1} y \, dy \equiv \int y^0 \cdot \tan y^{-1} \, dy \quad \text{→ Rewritten.}$$

$$f(n) = \tan y^{-1} \quad \text{and} \quad g(n) = y^0$$

$$= f(n) \int g(n) \, dy - \int \left(f'(n) \int g(n) \, dy \right) \, dy.$$

$$\tan y^{-1} \int y^0 \, dy - \int \left(\frac{d}{dy} \tan y^{-1} \int y^0 \, dy \right) \, dy.$$

$$\tan y^{-1} y^{0+1} - \int \left(\frac{1}{1+y^2} y^{0+1} \right) \, dy.$$

$$y \tan^{-1} y - \int \left(\frac{y}{1+y^2} \right) \, dy.$$

$$y \tan^{-1} y \quad \text{Multiple / Divide by "2":} \quad - \frac{1}{2} \int \left(\frac{2y}{1+y^2} \right) \, dy$$

$$= y \tan^{-1} y - \frac{1}{2} \int \frac{2y}{1+y^2} \, dy$$

$$y \tan^{-1} y - \frac{1}{2} \ln |1+y^2| + C$$

$$y \tan^{-1} y - \frac{1}{2} \ln (1+y^2) + C$$

$$y \tan^{-1} y - \ln \sqrt{1+y^2} + C$$

As we know
 $\int \frac{1}{n} \, dn = \ln |n| + C$
 if the derivative of
 Denominator is given in
 numerator, then we
 can write in the log
 form

Remember
 Always

See this

$\log (1+y^2)^{1/2}$
 we can write this, using log
 formulas.

$$\frac{1}{2} \ln (1+y^2)$$

Remember
 SEE
 (10K)

$$y \tan^{-1} y - \ln \sqrt{1+y^2} + C$$

Answer

$$12) \int \sin^{-1} y \, dy$$

Solution:-

$$\int y^0 \sin^{-1} y \, dy \quad \rightarrow \text{Rewritten}$$

$$\boxed{f(y) = y^0 \text{ and } g(y) = \sin^{-1} y}$$

Cancel...

$$f(y) = \sin^{-1} y \quad \text{and} \quad g(y) = y^0$$

$$= f(y) \int g(y) \, dy - \int \left(f'(y) \int g(y) \, dy \right) dy$$

$$= \sin^{-1} y \int y^0 \, dy - \int \left(\frac{d}{dy} (\sin^{-1} y) \cdot \int y^0 \, dy \right) dy$$

$$= \sin^{-1} y \cdot y^1 - \int \left(\frac{1}{\sqrt{1-y^2}} \cdot y \right) dy$$

$$y \sin^{-1} y - \int \frac{y}{\sqrt{1-y^2}} \, dy$$

$$y \sin^{-1} y - \int (y) (1-y^2)^{-1/2} \, dy$$

$$y \sin^{-1} y + \frac{1}{2} (-2y) (1-y^2)^{-1/2} \, dy$$

→ Multiple & Divide with -2

$$y \sin^{-1} y + \frac{1}{2} \frac{(1-y^2)^{1/2+1/1}}{-1/2+1/1} + C$$

$$y \sin^{-1} y + \frac{1}{2} \frac{(1-y^2)^{1/2}}{1/2} + C$$

$$\boxed{y \sin^{-1} y + \sqrt{1-y^2} + C} \quad \text{Answer}$$

$$\left[\begin{array}{l} y \sin^{-1} y \\ y \sin^{-1} y \end{array} \quad \begin{array}{l} y \frac{(1-y^2)^{-1/2+1/1}}{-1/2+1/1} \\ \frac{y^2}{2} \frac{\sqrt{1-y^2}}{1/2} \end{array} \right]$$

Practice
It's not
included

13) $\int n \sec^2 n \, dn$

Solution:-

$f(n) = n$ and $g(n) = \sec^2 n$

$$= f(n) \int g(n) \, dn - \int \left(f'(n) \cdot \int g(n) \, dn \right) \, dn$$

$$= n \int \sec^2 n \, dn - \int \left(\frac{d}{dn} (n) \cdot \int \sec^2 n \, dn \right) \, dn$$

$$= n \tan n - \int (1) \cdot \tan n \, dn$$

$$= n \tan n - \int \tan n \, dn$$

Multiplying $\ominus$

$$n \tan n + \int \frac{(-\sin n)}{\cos n} \, dn$$

$$n \tan n + \ln |\cos n| + C$$

Formula:

$$\therefore \frac{1}{n} = \ln |n| + C$$

Answer

14) $\int 4n \sec^2 2n \, dn$

Solution:-

$f(n) = 4n$ and $g(n) = \sec^2 2n$

$$f(n) \int g(n) \, dn - \int \left(f'(n) \cdot \int g(n) \, dn \right) \, dn$$

$$4n \int \sec^2 2n \, dn - \int \left(\frac{d}{dn} (4n) \cdot \int \sec^2 2n \, dn \right) \, dn$$

$\frac{2n \cdot \tan 2n}{2 \text{ Deriv}}$

$$- \int \left(4 \cdot \frac{\tan 2n}{2} \right) \, dn$$

$$2n \tan 2n - \int 2 \tan 2n \, dn$$

$\int \tan = \ln |\sec| + C$

$$2n \tan 2n - 2 \cdot \ln |\sec 2n| + C$$

$$2n \tan 2n - \ln |\sec 2n| + C$$

Answer

ILATE

$$15) \int n^3 e^n dn$$

Solution:

$$f(n) = n^3 \quad \text{and} \quad g(n) = e^n$$

$$= f(n) \int g(n) dn - \int (f'(n) \cdot \int g(n) dn) dn$$

$$= n^3 \int e^n dn - \int \left(\frac{d}{dn} (n^3) \cdot \int e^n dn \right) dn$$

$$= n^3 \cdot e^n - \int (3n^2 \cdot e^n) dn$$

$$= n^3 e^n - \int (3n^2 e^n) dn$$

$$= n^3 e^n - 3 \int n^2 e^n dn$$

again integration by parts

$$f(n) = n^2 \quad \text{and} \quad g(n) = e^n$$

$$n^3 e^n - 3 \left[f(n) \int g(n) dn - \int (f'(n) \int g(n) dn) dn \right]$$

$$n^3 e^n - 3 \left[n^2 \int e^n dn - \int \left(\frac{d}{dn} (n^2) \int e^n dn \right) dn \right]$$

$$n^3 e^n - 3 \left[n^2 e^n - \int (2n e^n) dn \right]$$

Multiple (3) ↑

$$= n^3 e^n - 3n^2 e^n + 3 \cdot 2 \int n e^n dn$$

$$= n^3 e^n - 3n^2 e^n + 6 \int n e^n dn$$

Again Integrate By parts

$$f(n) = n \quad g(n) = e^n$$

$$= n^3 e^n - 3n^2 e^n + 6 \left[f(n) \int g(n) dn - \int (f'(n) \cdot \int g(n) dn) dn \right]$$

$$n^3 e^n - 3n^2 e^n + 6 \cdot \left[n e^n - \int (1) e^n dn \right]$$

$$n^3 e^n - 3n^2 e^n + 6 \cdot \left[n e^n - e^n + c \right]$$

$$n^3 e^n - 3n^2 e^n + 6n e^n - 6e^n + c$$

$$\boxed{n^3 e^n - 3n^2 e^n + 6n e^n - 6e^n + c} \quad \text{Answer}$$

16) $\int p^4 e^{-p} dp$

Solution:

Let, $p^4 = f(p)$ and $g(p) = e^{-p}$

$$= f(p) \int g(p) dp - \int \left(f'(p) \cdot \int g(p) dp \right) dp$$

$$= p^4 \int e^{-p} dp - \int \left(\frac{d}{dp}(p^4) \cdot \int e^{-p} dp \right) dp$$

$$= p^4 e^{-p} - \int 4p^3 e^{-p} dp$$

$$= \frac{p^4 e^{-p}}{-1} + 4 \int \frac{p^3 e^{-p}}{-1} dp \quad \text{Again By parts}$$

$$f(p) = p^3 \quad g(p) = e^{-p}$$

$$= -p^4 e^{-p} + 4 \int p^3 e^{-p} dp$$

$$= -p^4 e^{-p} + 4 \left[f(p) \int g(p) dp - \int (f'(p) \int g(p) dp) dp \right]$$

$$= -p^4 e^{-p} + 4 \left[\int p^3 e^{-p} dp - \int 3p^2 e^{-p} dp \right]$$

$$= -p^4 e^{-p} + 4 \left[-p^3 e^{-p} + 3 \int p^2 e^{-p} dp \right] \quad \text{Again by parts}$$

$$f(p) = p^2 \quad g(p) = e^{-p}$$

$$= -p^4 e^{-p} - p^3 e^{-p} + 3 \left[f(p) \int g(p) dp - \int (f'(p) \int g(p) dp) dp \right]$$

$$= -p^4 e^{-p} - p^3 e^{-p} + 3 \left[\int p^2 e^{-p} dp - \int 2p e^{-p} dp \right]$$

$$= -p^4 e^{-p} - p^3 e^{-p} + 3 \left[-p^2 e^{-p} + 2 \int p e^{-p} dp \right]$$

Again By parts

Direct written:-

$$-p^4 e^{-p} - p^3 e^{-p} - 3p^2 e^{-p} + 6p e^{-p} - 6 e^{-p} + C$$

Answer

Without
Combine like
terms

if you want
you can

Note:-

also practice of
this question

Also this is
answery

$$17) \int (n^2 - 5n) e^n dn$$

Solution:-

$$\text{Let, } f(n) = n^2 - 5n \quad g(n) = e^n$$

$$\begin{aligned} &= f(n) \int g(n) dn - \int \left(f'(n) \cdot \int g(n) dn \right) dn \\ &= (n^2 - 5n) \int e^n dn - \int \left(\frac{d}{dn} (n^2 - 5n) \cdot \int e^n dn \right) dn \\ &= (n^2 - 5n) e^n - \int (2n - 5) e^n dn \end{aligned}$$

Again By part

$$f(n) = 2n - 5 \quad g(n) = e^n$$

$$\begin{aligned} &= (n^2 - 5n) e^n - \left[f(n) \int g(n) dn - \int \left(f'(n) \int g(n) dn \right) dn \right] \\ &= (n^2 - 5n) e^n - \left[2n - 5 \int e^n dn - \int (2) (e^n) dn \right] \\ &= (n^2 - 5n) e^n - \left[(2n - 5) e^n + \int 2 e^n dn \right] \end{aligned}$$

$$= (n^2 - 5n) e^n - (2n - 5) e^n + 2 e^n + C$$

$$\begin{aligned} &= (n^2 e^n - 5n e^n) - (2n e^n - 5 e^n) + 2 e^n + C \\ &= n^2 e^n - 5n e^n - 2n e^n + 5 e^n + 2 e^n + C \end{aligned}$$

$$= \boxed{n^2 e^n - 7n e^n + 7 e^n + C} \rightarrow \text{Answer}$$

Also Common = e^n

$$\boxed{e^n (n^2 - 7n + 7) + C} \rightarrow \text{Answer also}$$

ILATE.

$$18) \int (r^2 + r + 1) e^r dr$$

Solution:-

$$f(n) = (r^2 + r + 1) \quad \text{and} \quad g(n) = e^r$$

$$f(n) \int g(n) dr - \int \left(f'(n) \cdot \int g(n) dr \right) dr$$

$$(r^2 + r + 1) \int e^r dr - \int \left(\frac{d}{dr} (r^2 + r + 1) \cdot \int e^r dr \right) dr$$

$$(r^2 + r + 1) e^r - \int \left((2r + 1) (e^r) \right) dr$$

Again Integration
 $f(n) = 2r + 1$ $g(n) = e^r$ by parts.

$$(r^2 + r + 1) e^r - \left[f(n) \cdot \int g(n) dr - \int \left(f'(n) \cdot \int g(n) dr \right) dr \right]$$

$$(r^2 + r + 1) e^r - \left[(2r + 1) \int e^r dr - \int \left(\frac{d}{dr} (2) \cdot e^r \right) dr \right]$$

$$(r^2 + r + 1) e^r - \left[(2r + 1) e^r + 2 \int e^r dr \right]$$

just for this

$$(r^2 + r + 1) e^r \quad \left[- (2r + 1) e^r + 2 e^r + C \right]$$

Multiply Multiply

$$= r^2 e^r + r e^r + e^r - (2r e^r + e^r) + 2 e^r + C$$

$$r^2 e^r + r e^r + e^r - 2r e^r - e^r + 2 e^r + C$$

Common = e^r

$$\left[(r^2 + r + 1 - 2r - 1 + 2) e^r + C \right]$$

Answer.
E

$$19) \int n^5 e^n \, dn$$

Solution: $f(n) = n^5$ and $g(n) = e^n$

$$\begin{aligned} f(n) \int g(n) \, dn &= \int \left(f'(n) \cdot \int g(n) \, dn \right) \, dn \\ n^5 \int e^n \, dn &= \int \left(\frac{d}{dn} (n^5) \cdot \int e^n \, dn \right) \, dn \end{aligned}$$

$$= (n^5) e^n - \int (5n^4)(e^n) \, dn$$

$$= n^5 e^n - 5 \int n^4 e^n \, dn \rightarrow \text{Again by parts}$$

$f(n) = n^4$ $g(n) = e^n$
 $\hookrightarrow$ Suppose it's equation (1)

because of lengthy question.

$$\frac{n^5 e^n}{n^4 e^n} \rightarrow 5$$

$$\int n^4 e^n \, dn$$

$$n^4 \int e^n \, dn - \int 4n^3 \int e^n \, dn = n^4 e^n - 4 \int n^3 e^n \, dn$$